

Zhiyuan Fan

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Academic and Research Appointments

Harvard University

Postdoctoral Fellow in Power System Economics

John A. Paulson School of Engineering and Applied Sciences (SEAS)

Salata Institute for Climate and Sustainability

Mentors: Prof. Le Xie and Prof. James H. Stock

Cambridge, MA

2026–Present

Columbia University

Research Associate

Center on Global Energy Policy, School of International and Public Affairs (SIPA)

Supervisor: Dr. S. Julio Friedmann

New York, NY

2020–2022

Education

Columbia University

Ph.D. in Earth and Environmental Engineering

Advisor: Prof. Bolun Xu

Dissertation: *Strategic Climate Mitigation Decisions Across Policy Contexts*

New York, NY

2022–2026

Columbia University

M.S. in Mechanical Engineering, Energy Concentration

New York, NY

2018–2020

The Hong Kong University of Science and Technology

B.Eng. (Hons.) in Aerospace Engineering, Minor in Mathematics

Hong Kong

2014–2018

Research Interests

- **Energy Infrastructure and Long-Term Transition:** investment, deployment, and transformation of energy systems under decarbonization.
- **Incentive and Market Design:** mechanisms for long-term investment, reliability, flexibility, and climate mitigation.
- **Game Theory and Strategic Decision-Making:** strategic interactions among firms, nations, and other actors in energy systems.
- **Energy Economics and Policy:** economic drivers, policy interventions, and decisions under uncertainty in the energy transition.
- **Agentic AI for Economic Decision-Making:** how AI agents reshape strategic behavior, bidding, markets, and mechanism design.

Publications

Manuscripts

1. **Zhiyuan Fan**, Tianyi Lin, and Bolun Xu. *Climate Leadership and Policy Diffusion in a Trade-Mediated Network Game*.
2. **Zhiyuan Fan**, Tianyi Lin, and Bolun Xu. *Managing Voluntary Renewable Programs: Pricing, Revenue Allocation, and Renewable Expansion*.
3. Ning Qi*, **Zhiyuan Fan***, Yousuf Baker, Beini Lu, Anne-Sophie Corbeau, Dan Esposito, and Bolun Xu. *Energy Security Accelerates Synthetic-Fuel Adoption with Climate Co-Benefits*.
4. **Zhiyuan Fan** and Shen Wang. *A Behavioral-Economic Evaluation of LLM Bidding Agents in Electricity Markets*.
5. Ruonan Pi, **Zhiyuan Fan**, and Bolun Xu. *Reducing Steel Production Costs and Emissions through Grid-Responsive Electric Arc Furnace Scheduling*.

*Equal contribution.

Peer-Reviewed Journal Articles

1. **Zhiyuan Fan** and Bolun Xu. *Design Optimization and Global Impact Assessment of Solar-Thermal Direct Air Carbon Capture*. *Environmental Science & Technology*, 60(24): 17206–17215, 2026.
2. **Zhiyuan Fan**, Elizabeth Dentzer, James Glynn, David S. Goldberg, S. Julio Friedmann, and Bolun Xu. *Enhancing Profit and CO₂ Mitigation: Commercial Direct Air Capture Design and Operation with Power Market Volatility*. *Engineering*, 62: 283–295, 2026.
3. Ning Qi, Kaidi Huang, **Zhiyuan Fan**, and Bolun Xu. *Long-Term Energy Management for Microgrid with Hybrid Hydrogen-Battery Energy Storage: A Prediction-Free Coordinated Optimization Framework*. *Applied Energy*, 377: 124485, 2025.
4. Caleb M. Woodall, **Zhiyuan Fan**, Yushan Lou, Amar Bhardwaj, Amit Khatri, Mahak Agrawal, Colin F. McCormick, and S. Julio Friedmann. *Technology Options and Policy Design to Facilitate Decarbonization of Chemical Manufacturing*. *Joule*, 6(11): 2474–2499, 2022.
5. **Zhiyuan Fan** and S. Julio Friedmann. *Low-Carbon Production of Iron and Steel: Technology Options, Economic Assessment, and Policy*. *Joule*, 5(4): 829–862, 2021.

Reports, Roadmaps, and Policy Publications

1. David B. Sandalow, Colin McCormick, Alp Kucukelbir, S. Julio Friedmann, Trishna Nagrani, **Zhiyuan Fan**, Antoine M. Halff, et al. *Artificial Intelligence for Climate Change Mitigation Roadmap*. ICEF Innovation Roadmap Project, 2023.
2. Yushan Lou, Anne-Sophie Corbeau, and **Zhiyuan Fan**. *Quantifying Vented By-Product Hydrogen: A Case Study in China*. Oxford Institute for Energy Studies, 2023.
3. Yushan Lou, **Zhiyuan Fan**, S. Julio Friedmann, Anne-Sophie Corbeau, Mahak Agrawal, and Amit Khatri. *The Potential Role of Biohydrogen in Creating a Net-Zero World*. Center on Global Energy Policy, Columbia University, 2023.

4. David Sandalow, Roger Aines, **Zhiyuan Fan**, S. Julio Friedmann, Colin McCormick, Ann-Kathrin Merz, and Corinne Scown. *Low-Carbon Ammonia Roadmap*. ICEF Innovation Roadmap Project, 2022.
5. **Zhiyuan Fan**, Hadia Sheerazi, Amar Bhardwaj, Anne-Sophie Corbeau, Kathryn Longobardi, Adalberto Castaneda, Ann-Kathrin Merz, Caleb Woodall, Mahak Agrawal, Sebastian Orozco-Sanchez, and S. Julio Friedmann. *Hydrogen Leakage: A Potential Risk for the Hydrogen Economy*. Center on Global Energy Policy, Columbia University, 2022.
6. **Zhiyuan Fan**, Emeka Ochu, Sarah Braverman, Yushan Lou, Griffin Smith, Amar Bhardwaj, Jack Brouwer, Colin McCormick, and S. Julio Friedmann. *Green Hydrogen in a Circular Carbon Economy: Opportunities and Limits*. Center on Global Energy Policy, Columbia University, 2021.
7. S. Julio Friedmann, **Zhiyuan Fan**, Zachary Byrum, Emeka Ochu, Amar Bhardwaj, and Hadia Sheerazi. *Levelized Cost of Carbon Abatement: An Improved Cost-Assessment Methodology for a Net-Zero Emissions World*. Center on Global Energy Policy, Columbia University, 2020.
8. S. Julio Friedmann, Alex Zapantis, Brad Page, Chris Consoli, **Zhiyuan Fan**, Ian Havercroft, Harry Liu, Emeka Ochu, Nabeela Raji, Dominic Rassool, Hadia Sheerazi, and Alex Townsend. *Net-Zero and Geospheric Return: Actions Today for 2030 and Beyond*. Center on Global Energy Policy, Columbia University, 2020.
9. S. Julio Friedmann, **Zhiyuan Fan**, and Ke Tang. *Low-Carbon Heat Solutions for Heavy Industry: Sources, Options, and Costs Today*. Center on Global Energy Policy, Columbia University, 2019.

Selected Research Presentations

Decision-Making Across Policy Contexts in Climate Mitigation

- Andlinger Center for Energy and the Environment, Princeton University, Princeton, NJ. (2026)
- Power and AI Initiative, Harvard University, Cambridge, MA. (November 2025)

Optimization for Direct Air Capture System Operation and Design from Electric and Thermal Energy Perspectives

- 2025 INFORMS Annual Meeting, Atlanta, GA. (October 2025)

Selected Industry Experience

Carbon Direct (now Relae)

New York, NY

Summer Graduate Research Fellow

Summer 2024

- Conducted techno-economic and feasibility analyses for commercial decarbonization projects, including biohydrogen and carbon capture and storage (CCS) projects in Brazil and direct air capture (DAC) projects in the United States.
- Analyzed global low-grade iron ore supply and demand and associated decarbonization opportunities for the iron and steel sector.
- Performed technical due diligence on low-carbon technology companies, evaluating technical feasibility, performance claims, and key risks to support investment decisions.

Teaching and Mentoring

Harvard University

Teaching Assistant

- ES 215: Physical and Economical Operations of Sustainable Energy Systems, Fall 2026.

Columbia University

Teaching Assistant

- EAEE E4257: Environmental Data Analysis and Modeling, Spring 2025.
- ELEN E6906: Future Energy: Economics, Systems, Policies, Spring 2025.
- EAEE E4220: Energy System Economics and Optimization, Fall 2023.
- MECH E4320: Introduction to Combustion, Fall 2019.

Research Mentoring

- Ruonan (Pina) Pi, M.S. in Earth and Environmental Engineering, Columbia University, 2026.
Current position: Ph.D. student, University of Washington, Seattle.
- Beini Lu, M.S. student in Earth and Environmental Engineering, Columbia University.
Research mentee, 2025–Present.
- Elizabeth Dentzer, summer research mentee, Columbia University, 2023.
Current position: Ph.D. student in Mechanical and Aerospace Engineering, Princeton University.

Professional Service

Academic and Professional Activities

- Organizing Team, *Monthly Deep Dive Series: Critical Issues Facing the Grid*, Harvard University, 2026–Present.
- Session Chair, 2025 INFORMS Annual Meeting, Atlanta, GA.

Ad Hoc Reviewer

- *Nature Energy*; *Nature Sustainability*; *Nature Communications*; *Joule*; *Environmental Research Letters*; *IEEE Transactions on Smart Grid*; *IEEE Transactions on Sustainable Energy*; *IEEE Transactions on Energy Markets, Policy and Regulation*; NeurIPS.

Last updated: September 2026